Description

Cauliflower, *Brassica oleracea* var. *botrytis*, is an herbaceous annual or biennial vegetable plant in the family *Brassicaceae* grown for its edible head. The head is actually a mass of abortive flowers (flowers which are unable to produce fruit or seed as they possess only female reproductive organs; the male organs are either underdeveloped or totally lacking). Cauliflower plants are shallow rooted with a small, thickened stem. The ribbed leaves branch off the top of the stem and are light green in color. The plant can reach a height of 1–1.5 m (3.3–4.9 ft) and is most commonly grown as an annual, harvested between 60 and 100 days after planting. The origin of cauliflower has not been definitively determined but its ancestor, wild cabbage, is thought to have originated in ancient Asia Minor.

Uses

Cauliflower is grown for consumption as a vegetable, either fresh or cooked. May be processed for freezing or pickling.

Propagation

Basic requirements Cauliflower is a cool season crop and grows best in well draining, organic soil at a pH of 6.5 or above. A high amount of organic matter in the soil will help to hold moisture. The plant requires consistent cool temperatures to prevent 'buttoning' - the formation of several small heads instead of one large one. Cauliflower is less cold hardy than its relatives and it should be planted after the last frost as sub-freezing temperatures are likely to damage the plant. It is best grown as a Fall crop in cooler areas. Only mature plants will survive frost in the Fall. Transplants Cauliflower is most successfully grown from transplants started in the glasshouse or indoors. Peat pots work well for growing seedlings as damage to the roots during transplanting can be avoided. Plant seeds in a sterile seed starting mix to a depth of 0.6-1.3 cm (0.25-0.5 in) deep and keep moist. Seedlings can be transplanted when they are approximately 6 weeks old after hardening. Plant spacing will greatly affect the final size of the cauliflower head and the plants should be spaced at least 46 cm (18 in) apart leaving approximately 75 cm (30 in) between rows. General care and maintenance Cauliflower requires consistent moisture during the growing season in order to produce large, tender heads. Do not let the soil dry out. The plants also require ample nitrogen and this should b applied as a side-dressing of fertilizer about half way through the growing season. Cauliflower heads can be discolored by sunlight and when the cauliflower head has reached 2.5–5.0 cm (1–2 in) the plant requires 'blanching', a process that ensures that the head remains white. This is achieved by gathering the longest outer leaves together at the head and tying them with twine. Remove the twine periodically to check on the progress of the head, check for pests and to allow the head to dry out after rain. Self blanching varieties are available which grow leaves which naturally cover the head. Harvesting Cauliflower heads are ready for harvest when the head reaches 15 to 20 cm in diameter (6-8 in), usually about 7 to 12 days after blanching begins. Harvest the head by cutting the main stem with a sharp knife, including some of the central leaves which will protect the head. The heads can quickly become overmature, leading to reduced quality. Be sure to harvest the heads while they are still firm, before individual florets can be distinguished.

Common Pests and Diseases

Diseases

Category : Bacterial

Bacterial soft rot *Erwinia caratovora*

Symptoms

Water-soaked lesions on leaves and flower heads which expand to form a large rotted mass; surface of lesions usually crack and exude slimy liquid which turns tan, dark brown or black on exposure to air

Cause

Bacterium

Comments

Bacteria are easily spread on tools and by irrigation water; disease emergence favored by warm, moist conditions

Management

Chemical treatments are not available for bacterial soft rot, control relies on cultural practices; rotate crops; plant in well-draining soils or raised beds; only harvest heads when they are dry; avoid damaging heads during harvest

Blackleg *Leptosphaeria maculans*

Symptoms

Damping-off of seedlings; round or irregularly shaped gray necrotic lesions on leaves with dark margins; lesions may be covered in pink masses in favorable weather conditions

Cause

Fungus

Comments

Favors warm, wet conditions; higher temperatures result in the development of more visible symptoms

Management

Use disease free seed or treat with hot water to remove fungus prior to planting; remove and destroy crop debris after harvest or plow deeply into soil

Category : Fungal

Black rot *Xanthomonas campestris*

Symptoms

Irregularly shaped dull yellow areas along leaf margins which expand to leaf midrib and create a characterstic "V-shaped" lesion; lesions may coalesce along the leaf margin to give plant a scorched appearance

Cause

Bacterium

Comments

Pathogen is spread via infected seed or by splashing water and insect movement; disease emergence favored by warm and humid conditions

Management

Primary method of controlling black rot is through the use of good sanitation practices; rotate crops to non-cruciferous crops every 2 years; plant resistant varieties; control cruciferous weed species which may act as a reservoir for bacteria; plant pathogen-free seed

Clubroot *Plasmodiophora brassicae*

Symptoms

Slow growing, stunted plants; yellowish leaves which wilt during day and rejuvenate in part at night; swollen, distorted roots; extensive gall formation

Cause

Fungus

Comments

Can be difficult to distinguish from nematode damage; fungus can survive in soil for periods in excess of 10 years; can be spread by movement of contaminated soil and irrigation water to uninfected areas

Management

Once the pathogen is present in the soil it can survive for many years, elimination of the pathogen is economically unfeasible; rotating crops generally does not provide effective control; plant only certified seed and avoid field grown transplants unless produced in a fumigated bed; applying lime to the soil can reduce fungus sporulation

Downy mildew *Hyaloperonospora parasitica*

Symptoms

Small angular lesions on upper surface of leaves which enlarge into orange or yellow necrotic patches; white fluffy growth on undersides of leaves

Cause

Fungus

Comments

Disease emergence favored by cool, moist conditions

Management

Remove all crop debris after harvest; rotate with non-brassicas; it is possible to control downy mildew with the application of an appropriate fungicide

Powdery mildew *Erysiphe cruciferarum*

Symptoms

Small white patches on upper and lower leaf surfaces which may also show purple blotching; patches coalesce to form a dense powdery layer which coats the leaves; leaves become chlorotic and drop from plant

Cause

Fungus

Comments

Disease emergence favored by dry season, moderate temperatures, low humidity and low levels of rainfall

Management

Plant resistant varieties; rotate crops; remove all crop debris after harvest; remove weeds; avoid excessive application of nitrogen fertilizer which encourages powdery mildew growth; powdery mildew can be controled by application of sulfur sprays, dusts or vapors

Sclerotinia stem rot (White mold) *Sclerotinia sclerotiorum*

Symptoms

Irregular, necrotic gray lesions on leaves; white-gray leions on stems; reduced pod set; shattering seed pods

Cause

Fungus

Comments

Disease emergence favors moderate to cool temperatures and high humidity

Management

Rotate crop to non-hosts (e.g. cereals) for at least 3 years; control weeds; avoid dense growth by planting in adequately spaced rows; apply appropriate foliar fungicides

White rust *Albugo candida*

Symptoms

White pustules on cotyledons, leaves, stems and/or flowers which coalesce to form large areas of infection; leaves may roll and thicken

Cause

Fungus

Comments

Fungus can survive for long periods of time in dry conditions; disease spread by wind

Management

Rotate crops; plant only disease-free seed; apply appropriate fungicide if disease becomes a problem

Category : Viral

Cauliflower mosaic Cauliflower mosaic virus (CaMV)

Symptoms

Mosaic patterns on leaves; veinclearing and or veinbanding; stunted plant growth; reduced head size

Cause

Virus

Comments

Transmitted by various species of aphid including the cabbage aphid, and peach aphid

Management

Control cruciferous weeds which can act as a reservoir for the virus; control aphid populations on plants by applying an appropriate insecticide

Ring spot *Mycosphaerella brassicicola*

Symptoms

Small, purple spots surrounded by a ring of water-soaked tissue on leaves which mature to brown spots with olive green borders 1-2 cm across; spots may develop numerous fruiting bodies which give them a black appearance or develop a concentric pattern; heavily infected leaves may dry up and curl inwards

Cause

Fungus

Comments

Ring spot requires cool, moist conditions to survive; disease symptoms typically develop in the fall and the peak of the infection occurs in winter

Management

Refrain from planting in areas known to have had disease previously; rotate crop to non-brassicas; sanitize tools and equipment regularly; apply appropriate fungicide if disease is identified in crop

Category : Fungal, Oomycete

Wirestem (Damping-off) *Rhizoctonia solani*

Symptoms

Death of seedlings after germination; brown-red or black rot girdling stem; seedling may remain upright but stem is constricted and twisted (wirestem)

Cause

Fungus

Comments

Disease emergence in seedlings favored by cool temperatures

Management

Plant pathogen-free seed or transplants that have been produced in sterilized soil; apply fungicide to seed to kill off any fungi; shallow plant seeds or delay planting until soil warms

Pests

Category : Insects

Beet armyworm *Spodoptera exigua*

Symptoms

Singular, or closely grouped circular to irregularly shaped holes in foliage; heavy feeding by young larvae leads to skeletonized leaves; shallow, dry wounds on fruit; egg clusters of 50-150 eggs may be present on the leaves; egg clusters are covered in a whitish scale which gives the cluster a cottony or fuzzy appearance; young larvae are pale green to yellow in color while older larvae are generally darker green with a dark and light line running along the side of their body and a pink or yellow underside

Cause

Insect

Comments

Insect can go through 3–5 generations a year

Management

Organic methods of controlling the beet armyworm include biological control by natural enemies which parasitize the larvae and the application of *Bacillus thuringiensis*; there are chemicals available for commercial control but many that are available for the home garden do not provide adequate control of the larvae

Cabbage aphid *Brevicoryne brassicaea*

Symptoms

Large populations can cause stunted growth or even plant death; insects may be visible on the plant leaves and are small, grey-green in color and soft bodied and are covered with a white waxy coating; prefer to feed deep down in cabbage head and may be obscured by the leaves

Cause

Insect

Comments

Cabbage aphids feed only on cruciferous plants but may survive on related weed species

Management

If aphid population is limited to just a few leaves or shoots then the infestation can be pruned out to provide control; check transplants for aphids before planting; use tolerant varieties if available; reflective mulches such as silver colored plastic can deter aphids from feeding on plants; sturdy plants can be sprayed with a strong jet of water to knock aphids from leaves; insecticides are generally only required to treat aphids if the infestation is very high - plants generally tolerate low and medium level infestation; insecticidal soaps or oils such as neem or canola oil are usually the best method of control; always check the labels of the products for specific usage guidelines prior to use

Cabbage looper *Trichoplusia ni*

Symptoms

Large or small holes in leaves; damage often extensive; caterpillars are pale green with a white lines running down either side of their body; caterpillars are easily distinguished by the way they arch their body when moving; eggs are laid singly, usually on the lower leaf surface close to the leaf margin, and are white or pale green in color

Cause

Insect

Comments

Cabbage looper can be identified by their characteristic "looping" movement in which they arch their body and bring the back portion of the body forward to meet the front

Management

Looper populations are usually held in check by natural enemies; if they do become problematic larvae can be hand-picked from the plants; biological controls such as spraying with *Bacillus thuringiensis* can be effective at controlling looper numbers; application of appropriate insecticide also controls looper populations; selective insecticides help to protect populations of natural enemies on crop

Cucumber beetles (Western spotted cucumber beetle) *Diabrotica undecimpunctata*

Symptoms

Stunted seedlings; damaged leaves, stems and/or petioles

Cause

Insect

Comments

Beetles overwinter in soil and leaf litter

Management

Monitor new planting regularly for signs of beetle; apply appropriate insecticides

Cutworms *Agrotis* spp.

Peridroma saucia

Nephelodes minians

and others

Symptoms

Stems of young transplants or seedlings may be severed at soil line; if infection occurs later, irregular holes are eaten into the surface of fruits; larvae causing the damage are usually active at night and hide during the day in the soil at the base of the plants or in plant debris of toppled plant; larvae are 2.5–5.0 cm (1–2 in) in length; larvae may exhibit a variety of patterns and coloration but will usually curl up into a C-shape when disturbed

Cause
Insect

Comments
Cutworms have a wide host range and attack vegetables including asparagus, bean, cabbage and other crucifers, carrot, celery, corn, lettuce, pea, pepper, potato and tomato

Management
Remove all plant residue from soil after harvest or at least two weeks before planting, this is especially important if the previous crop was another host such as alfalfa, beans or a leguminous cover crop; plastic or foil collars fitted around plant stems to cover the bottom 3 inches above the soil line and extending a couple of inches into the soil can prevent larvae severing plants; hand-pick larvae after dark; spread diatomaceous earth around the base of the plants (this creates a sharp barrier that will cut the insects if they try and crawl over it); apply appropriate insecticides to infested areas of garden or field if not growing organically

Diamondback moth *Plutella xylostella*

Symptoms
Young larvae feed between upper and lower leaf surface and may be visible when they emerge from small holes on the underside of the leaf; older larvae leave large, irregularly shaped shotholes on leaf undersides, may leave the upper surface intact; larvae may drop from the plant on silk threads if the leaf is disturbed; larvae are small (1 cm/0.3 in) and tapered at both ends; larvae have to prolegs at the rear end that are arranged in a distinctive V-shape

Cause
Insect

Comments
Larvae take between 10 and 14 days to mature and spin a loose, gauze-like cocoon on leaves or stems to pupate

Management
Larvae can be controlled organically by applications of *Bacillus thurengiensis* or Entrust; application of appropriate chemical insecticide is only necessary if larvae are damaging the growing tips of the plants

Flea beetles *Phyllotreta cruciferae*

Symptoms

Small holes or pits in leaves that give the foliage a characteristic “shothole” appearance; young plants and seedlings are particularly susceptible; plant growth may be reduced; if damage is severe the plant may be killed; the pest responsible for the damage is a small (1.5–3.0 mm) dark colored beetle which jumps when disturbed; the beetles are often shiny in appearance

Cause

Insects

Comments

Flea beetles may overwinter on nearby weed species, in plant debris or in the soil; insects may go through a second or third generation in one year

Management

In areas where flea beetles are a problem, floating row covers may have to be used prior to the emergence of the beetles to provide a physical barrier to protect young plants; plant seeds early to allow establishment before the beetles become a problem - mature plants are less susceptible to damage; trap crops may provide a measure of control - cruciferous plants are best; application of a thick layer of mulch may help prevent beetles reaching surface; application on diamotecoeus earth or oils such as neem oil are effective control methods for organic growers; application of insecticides containing carbaryl, spinosad, bifenthrin and permethrin can provide adequate control of beetles for up to a week but will need reapplied

Large cabbage white (Cabbageworm) *Pieres rapae*

ymptoms

Large ragged holes in leaves or bored into head; green-brown frass (insect feces) on leaves; caterpillar is green in color and hairy, with a velvet-like appearance; may have faint yellow to orange stripes down back; slow-moving compared with other caterpillars

Cause

Insect

Comments

Larvae cause damage by feeding on plants; can be distinguished from other caterpillars by its sluggish movement; in large numbers larvae can cause extensive damage very quickly

Management
Plant cHand-pick caterpillars from plants and destroy; scrape eggs from leaves prior to hatching; apply appropriate insecticide if infestation is very heavy

Thrips (Western flower thrips, Onion thrips, etc.) *Frankliniella occidentalis*

Thrips tabaci

Symptoms
If population is high leaves may be distorted; leaves are covered in coarse stippling and may appear silvery; leaves speckled with black feces; insect is small (1.5 mm) and slender and best viewed using a hand lens; adult thrips are pale yellow to light brown and the nymphs are smaller and lighter in color

Cause
Insect

Comments
Transmit viruses such as Tomato spotted wilt virus; once acquired, the insect retains the ability to transmit the virus for the remainder of its life

Management
Avoid planting next to onions, garlic or cereals where very large numbers of thrips can build up; use reflective mulches early in growing season to deter thrips; apply appropriate insecticide if thrips become problematic

Category : Nematodes

Root knot nematode *Meloidogyne* spp.

Symptoms
Galls on roots which can be up to 3.3 cm (1 in) in diameter but are usually smaller; reduction in plant vigor; yellowing plants which wilt in hot weather

Cause
Nematode

Comments

Galls can appear as quickly as a month prior to planting; nematodes prefer sandy soils and damage in areas of field or garden with this type of soil is most likely

Management

Plant resistant varieties if nematodes are known to be present in the soil ;check roots of plants mid-season or sooner if symptoms indicate nematodes; solarizing soil can reduce nematode populations in the soil and levels of inoculum of many other pathogens